

# Revised Generation Interconnection System Impact Study Report

## For Queue Project Q15-0431

Pico 138 kV  
March 15, 2016

### 1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q15-0431. The Interconnection Customer proposes a 118.2 MW solar photovoltaic facility interconnecting at the Pico 138 kV substation in the Pennsylvania area. The request was received on February 07, 2015. The study evaluated system intact and single contingency conditions using the regional planning case, including 250 higher-queued active requests and their assigned network upgrades.

### 2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 7 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$8,462,000.

### 3. Network Impacts

#### 3.1 Generator Deliverability and Single Contingency Analysis

| Monitored Facility                      | Contingency                                                   | Loading (%) | Rating (MVA) | DFAX (%) |
|-----------------------------------------|---------------------------------------------------------------|-------------|--------------|----------|
| Dorchester-Cecil Field 138kV            | Marcus Hook Energy Lp Substation-Tombstone Substation 500kV L | 119.4       | 269          | 8.9      |
| Cecil Field-Plymouth Meeting 138kV L    | Tombstone-Marcus Hook Energy Lp 500kV line                    | 119.3       | 280          | 8.9      |
| Fruitland-St. Johns 138kV               | Tombstone-Marcus Hook Energy Lp 500kV                         | 117.2       | 173          | 8.9      |
| Barbadoes-Plymouth Meeting 138kV L      | Marcus Hook Energy Lp-Tombstone 500kV                         | 116.3       | 300          | 8.9      |
| Dorchester-Hathaway St Substation 138kV | Marcus Hook Energy Lp-Tombstone 500kV                         | 113.8       | 267          | 8.9      |
| St. Johns-Hathaway Street 138kV         | Tombstone-Marcus Hook Energy Lp 500kV                         | 110.9       | 214          | 8.9      |
| Pico-Stent 138kV                        | Big Swan-Owatonna Substation 345kV                            | 101.8       | 191          | 42.2     |

#### 3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

#### 3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

### 4. Required Network Upgrades and Cost Estimates

| Facility                                | Required Upgrade                            | Cost Estimate | Allocation to Project |
|-----------------------------------------|---------------------------------------------|---------------|-----------------------|
| Dorchester-Cecil Field 138kV            | Rebuild line with higher-capacity conductor | \$2,612,000   | 58% (\$1,505,000)     |
| Cecil Field-Plymouth Meeting 138kV L    | Rebuild line with higher-capacity conductor | \$2,179,000   | 58% (\$1,255,000)     |
| Fuitland Substation-St. Johns 138kV     | Rebuild line with higher-capacity conductor | \$2,213,000   | 58% (\$1,275,000)     |
| Barbadoes-Plymouth Meeting 138kV line   | Rebuild line with higher-capacity conductor | \$3,182,000   | 58% (\$1,833,000)     |
| Dorchester-Hathaway St Substation 138kV | Rebuild line with higher-capacity conductor | \$1,585,000   | 58% (\$913,000)       |
| Hathaway St-St. Johns 138kV L           | Rebuild line with higher-capacity conductor | \$2,430,000   | 58% (\$1,399,000)     |
| Pico Substation-Stent 138kV line        | Reconductor line                            | \$505,000     | 56% (\$282,000)       |
|                                         | Total allocated to project                  |               | \$8,462,000           |

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.